

Combining soil and tree-stem flux measurements and soil gas profiles to understand CH₄ pathways in *Fagus sylvatica* forests

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Abstract

Quantifying and understanding fluxes of methane (CH₄) and carbon dioxide (CO₂) in natural soil–plant–atmosphere systems are crucial to predict global climate change. Wetland herbaceous species or tree species at waterlogged sites are known to emit large amounts of CH₄. Upland forest soils are regarded as CH₄ sinks and tree species like upland beech are not known to significantly emit CH₄. Yet, data are scarce and this assumption needs to be tested.

We combined measurements of soil–atmosphere and stem–atmosphere fluxes of CO₂ and CH₄, and soil gas profiles to assess the contribution of the different ecosystem compartments at two upland beech forest sites in Central Europe in a case study. Soil was a net CH₄ sink at both sites, though emissions were detected consistently from beech stems at one site. Although stem emissions from beech stems were high compared to known fluxes from other upland tree species, they were substantially lower compared to the strong CH₄ sink of the soil. Yet, we observed extraordinarily large CH₄ emissions from one beech tree that was 140% of the CH₄ sink of the soil. The soil gas profile at this tree indicated CH₄ production at a soil depth > 0.3 m, despite the net uptake of CH₄ consistently observed at the soil surface. Field soil assessment showed strong redoximorphic color patterns in the adjacent soil and supports this evaluation. We hypothesize that there is a transport link between the soil and stem via the root system representing a preferential transport mechanism for CH₄ despite the fact that beech roots usually do not bear aerenchyma. The high mobility of gases requires a holistic view on the soil–plant–atmosphere system. Therefore, we recommend including field soil assessment and soil gas profiles measurements when investigating soil–atmosphere and stem–atmosphere fluxes to better understand the sources of gases and their transport mechanisms.

Key words: CH₄ / soil gas profile / gas flux / stem gas flux / CO₂ / methanogenesis

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1 Introduction

Carbon dioxide (CO₂) and methane (CH₄) are the most important greenhouse gases co-responsible for the global climate change. Terrestrial ecosystems like forests play a major role in global carbon fluxes, where the most active and important surfaces are thought to be the soil surface and foliage. Yet there is evidence that the stem surface can also play a substantial role at the ecosystem level as observed in temporarily flooded tropical forests (Pangala et al., 2013), temperate wetland forests (Pangala et al., 2015; Terazawa et al., 2015), and wet boreal forests (Machacova et al., 2016). The sources of CH₄ emitted by the stem surface can be diverse (Mukhin and Voronin, 2011; Covey et al., 2012; Pangala et al., 2013; Wang et al., 2016) and are mostly unclear. Emissions may result from either microbial or plant physiological gas production within the stem where elevated concentration of CH₄ and CO₂ can be found (Mukhin and Voronin, 2011; Covey et al., 2012). Furthermore, gas may also be produced within the soil and transported by roots into above-ground plant tissues.

The presence and transport of CH₄ in tree stems is known for decades (Bushong, 1907), but most studies focused on wet or temporarily flooded sites where usually high CH₄ emissions can be detected. In contrast to wetlands, upland forests are considered to be CH₄ sinks due to strong CH₄ oxidation in upland forest soils where microbial methanotrophy dominates over methanogenesis (Le Mer and Roger, 2001; Smith et al., 2003). Upland tree species are thought to be a negligible source of CH₄, and are therefore not considered in current CH₄ flux inventories. Yet, recent studies suggest that stem–atmosphere flux might also be an overlooked important pathway of CH₄ in the upland forests (Megonigal and Guenther, 2008). European beech (*Fagus sylvatica* L.) is the most important broadleaved tree species in upland forests of Central Europe, which covers a wide range of soil and climatic conditions. Yet information about CH₄ fluxes in mature beech is lacking.

Therefore, we investigated whether stem fluxes of CH₄ and CO₂ can play a significant role compared to the soil–atmosphere fluxes in upland forests of mature beech. We included

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field soil assessment and soil gas profiles to estimate the potential and actual production of CH₄ and CO₂ within the soil in addition to the measured soil–atmosphere flux.

In this paper we introduce a combination of methods to understand pathways of soil gases and present first results from a case study from two mature beech mountain forest sites in Central Europe.

2 Material and methods

Measurements for this case study were carried out in two mature beech forests (> 100 y old) in Central Europe (Tab. 1). The Conventwald site is located in the Western Black Forest, Germany (48.02°N, 7.96°E, 800 m asl). The mean annual temperature is 6.6°C and mean annual precipitation 1400 mm (von Wilpert et al., 2000). The stand is dominated by 70% beech and 30% Silver fir (*Abies alba* L.). The Stitna site is a pure beech stand and is located in the White Carpathians near Stitna nad Vlari, Czech Republic (49.02°N, 17.58°E, at 560 m asl) (Darenova et al., 2016). The mean annual temperature is 7.5°C and mean annual precipitation 800 mm. The soils at both sites are Cambisols, yet soil pH, texture, and structure are starkly different (Tab. 1). At Stitna site, soil texture is loam (no coarse soil) and structure is blocky. The dry soil showed deep (> 0.2 m) vertical cracks. Redoximorphic color patterns were found at some locations. At the Conventwald site, texture is sandy loam with 30–60% coarse soil fraction. Soil structure is granular, and no redoximorphic color patterns were found.

Measurements of stem–atmosphere fluxes, soil–atmosphere fluxes and soil gas profiles were carried out during measurement campaigns from June 24–29, 2015, at the Conventwald site and from July 07–10, 2015, at the Stitna site. Trees, understorey vegetation and soil were mapped. We used a soil probe (0.025 m diameter, type Pürckhauer) for a systematic overview and a soil core sampler (0.08 m diameter, type Murauch) for detailed morphological soil analyses. Weather conditions were stable during the campaigns. Mean air temperature (T_A) was 16.8°C at Conventwald, mean soil water content at 0.3 m depth (SWC) was 22 vol%, which can be considered as moderately dry. Mean soil temperature at 0.3 m depth (T_S) was 9.6°C. The plot seemed to be homogeneous with almost no herbal species. In Stitna, T_A was 19.2°C, SWC was 26%, and T_S was 15.2°C at 0.3 m depth. Lab analysis of soil cores

showed that the water potential was lower at the Stitna site, i.e., the soil had less plant available water which is due to the higher clay content at the Stitna site. The topsoil even had exposed shrinkage cracks. The plot was rather homogeneous with almost no herbal species. An area with small exemplars of *Impatiens noli-tangere* (L.) indicated generally high soil moisture. Field soil assessment using a soil probe and a core sampler showed strong redoximorphic color patterns in this area.

Five representative trees per site were selected for the measurement of stem–atmosphere fluxes. Both CH₄ and CO₂ fluxes were simultaneously and repeatedly measured at the base of major roots and three stem heights (0.4, 1.2 and 2.0 m above soil) per tree using non-steady-state flow-through chamber systems. Three to four box chambers were installed per stem height and interconnected with tubes and a vacuum pump into one system (Machacova et al., 2014). Gas samples were taken at 0–30–60–90–130–170–210–260–310–360 min after chamber closure, stored in gas-tight vials (Labco, Lampeter, UK), and analysed for CH₄ and CO₂ concentrations with a gas chromatograph Tracera (Shimadzu, Kyoto, Japan) equipped with a barrier discharge ionization detector (BID) and Shin Carbon ST Micro column (Restek, Bellefonte, PA, USA). The BID, operating at 250°C, was supplied with helium (80 mL min^{−1}), used also as a carrier gas (15 mL min^{−1}). Oven temperature was kept at 70°C for 14 min and then increased to 200°C for 1.5 min. The gas samples were injected by an autosampler GX-271 (Gilson, Middleton, WI, USA). The CH₄ and CO₂ concentrations and the fluxes were calculated based on four-point calibration curves and linear least square fits of time series of the concentrations (Machacova et al., 2016).

Soil–atmosphere gas fluxes were measured using non-steady state flow-through chambers. The plot was divided regularly into subplots: the chambers' positions within these subplots were based on a randomised design. Soil–atmosphere fluxes were measured at 27 positions in Conventwald and 35 positions in Stitna. The chambers were installed one day before the measurement. The chamber base consisted of a PVC cylinder (diameter 0.17 m, height 0.25 m) and a mobile lid with a vent (Schack-Kirchner et al., 2001). The chamber air was circulated via tubes through a Greenhouse Gas Analyzer (GGA, Ultraportable, Los Gatos Research, San Jose, CA, US) for the analyses of CH₄ and CO₂ concentrations. The chambers were closed for the flux measurement. CH₄

Table 1: Description of the experimental sites.

Site	Conventwald ^a	Stitna ^b
Annual Precipitation / Temperature	1400 mm / 6.6°C	800 mm / 7.5°C
Soil type (FAO, 2006)	Hyperdistric Skeletic Cambisol	Eutric (Stagnic) Cambisol
Humus type / soil pH	More-like Moder / 3.5	Mull / 7.0
Texture / coarse soil fraction	Sandy loam / 30–60%	Loam / 0%
Stand height / diameter at 1.3 m height	29.9 m / 0.44 m	32.2 m / 0.35 m

^avon Wilpert et al. (2000);

^bDarenova et al. (2016).

and CO₂ fluxes were calculated linearly based on the gas concentration changes over time of the first four minutes.

Soil gas profiles were measured around two of the five selected trees using passive gas samplers (Laemmel et al., 2017), where the air in the sampling tube equilibrated over time with the adjacent soil air. The gas samplers were installed into predrilled holes (0.4 m depth) in a transect from the respective tree stem at a distance of 1, 2, and 4 m at the beginning of the campaign. The same GGA and automatic valve system were used as for the soil flux chambers.

The stem surface area per ha was conservatively estimated to compare and upscale the contribution of the soil–atmosphere and stem–atmosphere fluxes. Assuming cone shaped stems of 0.4 m diameter at breast height, 30 m height and a tree density of 300 trees per hectare yielded a stem surface area per ha that equals roughly 60% the soil surface area.

3 Results and discussion

Soil was a net source of CO₂ at all measured positions at both sites. Mean soil CO₂ efflux ($\pm$ standard deviation) was $3.9 \pm 2.3 \mu\text{mol m}^{-2} \text{s}^{-1}$ in Conventwald and $6.8 \pm 2.3 \mu\text{mol m}^{-2} \text{s}^{-1}$ in Stitna. All soil CO₂ profiles showed increasing concentrations with depth that reached up to 12,000 ppm at depth > 0.4 m at both sites, indicating soil CO₂ production throughout the profile. Tree stems were also sources of CO₂ at both sites, with emissions of 0.8 ± 0.6 and $1.5 \pm 0.9 \mu\text{mol m}^{-2} \text{s}^{-1}$ in Conventwald and Stitna, respectively. Stem–atmosphere fluxes (per m² stem surface) of CO₂ reached 20–22% of the soil–atmosphere fluxes (per m² soil surface) at both sites and are in a good agreement with previous findings (Janssens et al., 2001; Damesin et al., 2002).

Soil was a net sink for CH₄ at all measured positions at both sites reported here as a negative flux (Fig. 1). Mean soil CH₄ fluxes

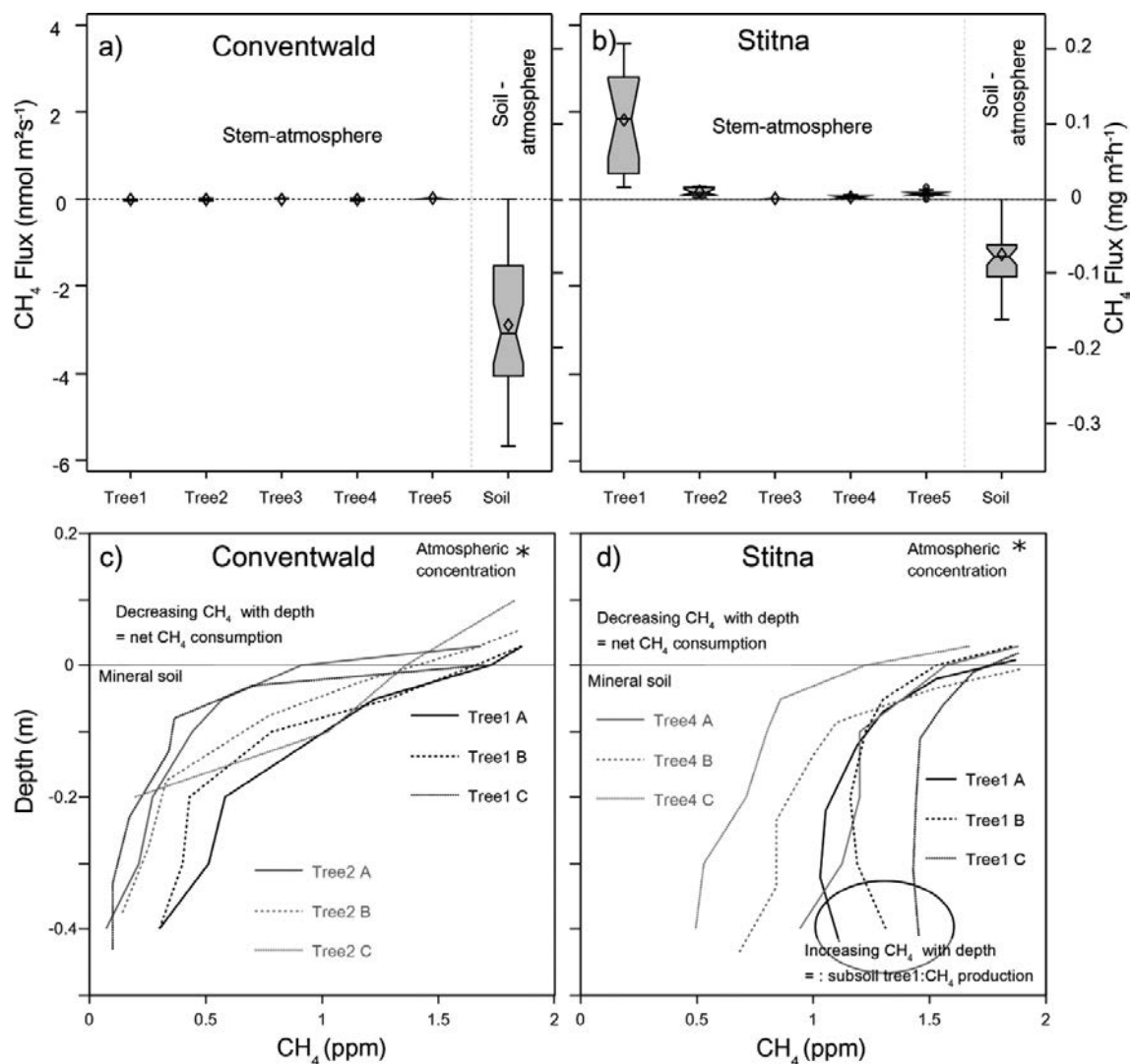


Figure 1: (a) Stem and soil CH₄ flux at Conventwald. Soil is a strong CH₄ sink, stem emissions are minimal (flux unit is per m² stem area and soil, respectively; please note the different units at the left and right axis). (b) At Stitna, *Tree1* emits large amounts of CH₄, while the other trees have only low emissions. (c) Soil CH₄ profiles at *Tree1* (black) and *Tree2* (grey) at Conventwald; profile A is next to the tree, then B and C. CH₄ concentrations decrease with soil depth indicating CH₄ consumption throughout the profile. (d) Soil CH₄ concentrations at Stitna are also sub-ambient, but not as low as at Conventwald. Soil CH₄ profiles at *Tree1* show a slight increase between 0.3 and 0.4 m, indicating local CH₄ production.

were $-2.9 \pm 1.7 \text{ nmol m}^{-2} \text{ s}^{-1}$ in Conventwald and $-1.3 \pm 0.8 \text{ nmol m}^{-2} \text{ s}^{-1}$ in Stitna (Fig. 1a, b). The observed soil CH_4 uptake is in accordance with the soil conditions, specifically the low SWC at both sites at time of sampling (Smith et al., 2003). Soil CH_4 concentrations were sub-ambient (Fig. 1c, d), indicating net CH_4 oxidation within the soil. All but two soil gas profiles showed decreasing CH_4 concentrations with depth throughout the soil profile (Fig. 1c, d). The two CH_4 profiles closest to *Tree1* in Stitna showed an increase of CH_4 between 0.3–0.4 m deep, indicating local CH_4 production at that depth. The location near *Tree1* also had a higher SWC compared to the rest of the experimental plot as detected and as indicated by the mapped *I. noli-tangere*. Additionally, we found strong redoximorphic color patterns in the soil, indicating temporally anaerobic conditions that are suitable for CH_4 production. Therefore, it is not surprising to find a local hot spot of CH_4 production in the deeper adjacent soil of *Tree1*. The CH_4 produced there, however, was re-consumed in the overlying aerated soil and had no effect on the CH_4 uptake at the soil surface, which was not different from the rest of the experimental plot. Hence, we might expect both low and high affinity methanotrophic microbes present in the soil profile.

Stem fluxes of CH_4 were not significant in Conventwald and negligible compared to the high CH_4 uptake by the soil (Fig. 1a). At Stitna, however, the beech trees consistently emitted CH_4 into the atmosphere ($0.52 \pm 0.92 \text{ nmol m}^{-2} \text{ s}^{-1}$; Fig. 1). While the mean CH_4 emissions per m^2 stem surface of four trees at Stitna were between 0.01 and $0.18 \text{ nmol m}^{-2} \text{ s}^{-1}$, the CH_4 emissions of *Tree1* amounted up to $1.8 \text{ nmol m}^{-2} \text{ s}^{-1}$. Moreover, the CH_4 emissions of *Tree1* tended to increase with stem height (1.1, 0.3, 2.5, and $3.4 \text{ nmol m}^{-2} \text{ s}^{-1}$ from the root base to 2 m stem height). This is surprising since other studies found a decrease with stem height (Rusch and Rennenberg, 1998; Terazawa et al., 2007; Pangala et al., 2013; Kutschera et al., 2016). Yet, the pattern of increase with stem height is not clear and difficult to interpret. Factors such as the density of lenticels might also have an effect and should be considered.

Our results imply that the soil CH_4 production detected only in the vicinity of *Tree1* was responsible for the substantial CH_4 stem emissions of *Tree1* at Stitna (Fig. 2). The observed CH_4 concentration in the subsoil was still sub-ambient (Fig. 1d), yet the increase with depth is a clear indication of CH_4 production. We suppose that CH_4 concentrations closer to *Tree1* or deeper than 0.4 m further increased, and thus exceeded

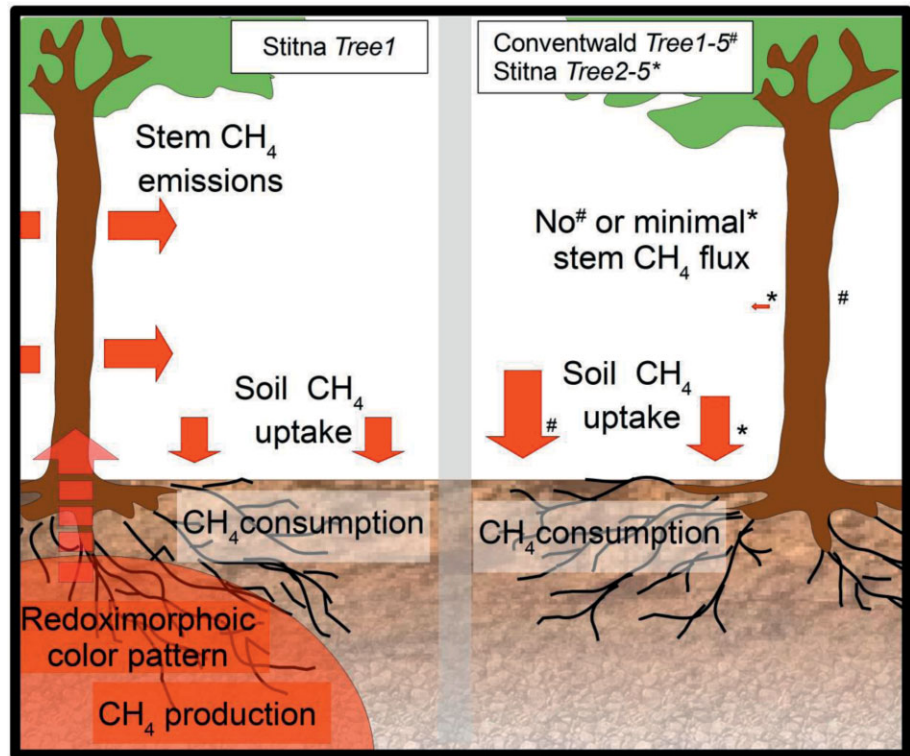


Figure 2: *Tree1* at Stitna (left) showed substantial CH_4 stem emissions in contrast the other investigated trees (right). The subsoil around *Tree1* at Stitna (left) showed strong redoximorphic color patterning and CH_4 production in the subsoil which we hypothesize is the source of the CH_4 emitted by the stem.

2 ppm. The transport mechanism of soil CH_4 into the tree roots and further up the tree could be either via dissolved CH_4 in the xylem or via gas transport in intercellular spaces (Machacova et al., 2013; Kutschera et al., 2016). The mechanism is unclear though and needs to be clarified.

In our limited data set we are lacking, among others, the contribution of the foliage and seasonal variability. Nevertheless it can be used for a first estimation of how soil and stems contribute to CH_4 fluxes at the scale of forest ecosystems. While CH_4 stem fluxes were negligible at Conventwald, they represented a substantial contribution at Stitna. The CH_4 emissions of *Tree1* dominated the observed stem fluxes, yet also the lower fluxes of *Tree2–5* at Stitna were substantial compared to fluxes from upland tree species, e.g., Scots pine (Machacova et al., 2016). Due to the limited number of trees and soil gas profiles, it is hardly possible to estimate the precise contribution of the different fluxes from these compartments to the ecosystem CH_4 exchange, especially since we lack information regarding the heterogeneity of tree methane emissions. Considering *Tree1* individually and scaling the stem flux per m^2 stem surface to flux per m^2 soil surface, the CH_4 uptake of the surrounding soil is completely offset by the strong emissions of *Tree1*. Hence, depending on the number of trees emitting CH_4 and the intensity of the respective emissions, the CH_4 uptake of the site could be significantly reduced or even offset when considering stem emissions.

For a more reliable estimation more trees and also stem flux measurements above 2 m would be necessary (Machacova et al., 2016). Likewise, a better allometric function describing the stem surface area would further improve the tree flux estimation. This short measurement campaign only obtained an initial snapshot during a period when soil moisture was low and thus CH₄ production was probably lower than normal. Repeated measurements are necessary as less uptake of CH₄ by the topsoil and more CH₄ production in the subsoil are probable. If so, real CH₄ stem emissions, e.g., when the soil moisture is higher, may be even more relevant than seen in this short campaign.

4 Conclusion

We observed for the first time that mature beech trees in an upland forest can emit CH₄ from their stems. The presented combination of methods allowed for direct identification of CH₄ production in the subsoil and linking it to stem fluxes, and thus helps to understand CH₄ pathways in *Fagus sylvatica* forests. As known from studies on carbon fluxes in the plant–soil system (Högberg and Read, 2006) we want to promote a holistic view on the soil–plant–atmosphere system and recommend including field methods and measurement of soil gas profiles.

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